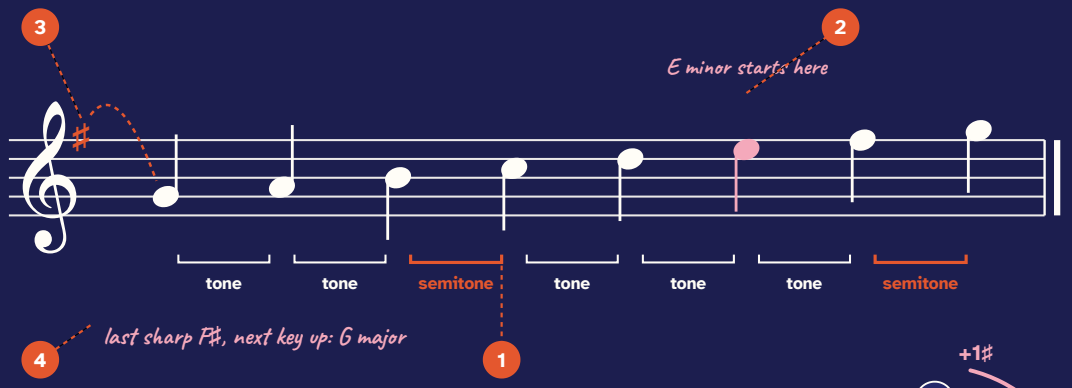
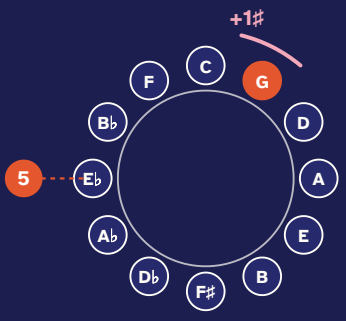
PART IV · NOTES IN FAMILIES

# Scales and keys

Notes travel in families. Build the scales that whole pieces are made from, then learn to read the key a piece lives in: its signature, its home note, its place on the circle. All of it is in one scale.




1 **Major scales** p. 93
T T S T T T S: the two short steps

2 **Minor scales** p. 96
the major with 3, 6 and 7 one key lower

3 **Keys and key signatures** p. 99
the sharps at the start apply all the way

4 **Finding the key** p. 102
last sharp, next key up; second-last flat

5 **The circle of fifths** p. 105
go right: one more sharp each step

★ **Tips: when a note sounds wrong** p. 108

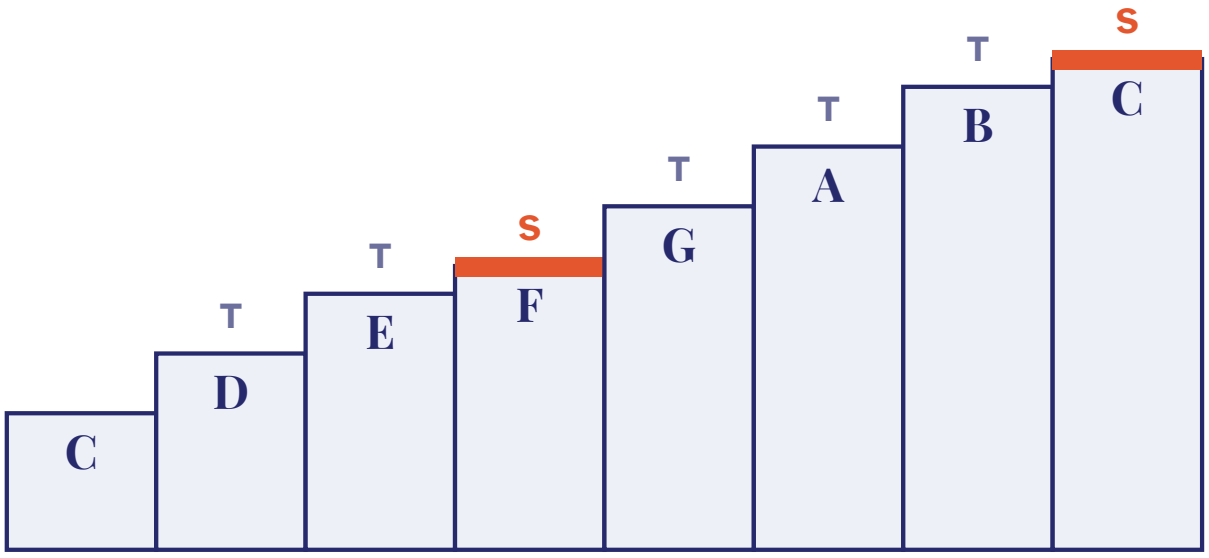
PART IV · SCALES AND KEYS · 1

# Major scales

A **scale** climbs from a home note, the **key note**, to the next note with the same name, using every letter once. A **major** scale always takes the same steps, whatever note it starts on.

THE SECRET

**Every major scale: tone, tone, semitone, tone, tone, tone, semitone.**



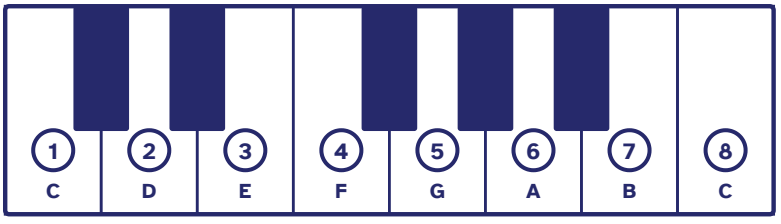

*Tone, tone, semitone, tone, tone, tone, semitone*

THE RULE

**T = tone = one key between. S = semitone = no key between: the very next key, best friends. Every major scale: T T S T T T S.**

EXAMPLE 1

**Where are the short steps?**




?

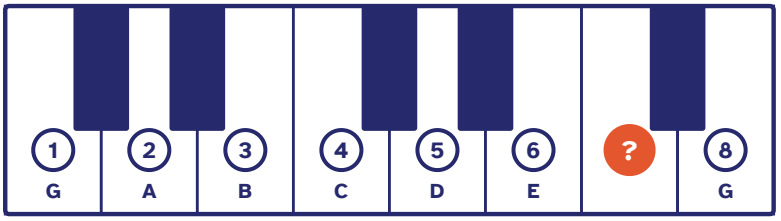
1. **Play C up to C** on the white keys, numbering the notes 1 to 8.
2. **Between each pair,** look for a black key. A key between means a tone.
3. **Which pairs have no black key** between them?

Only **3–4** (E–F) and **7–8** (B–C): nothing between. Those are the two short steps of every major scale.

PART IV · SCALES AND KEYS · 1 · CONTINUED

EXAMPLE 2

### Build G major

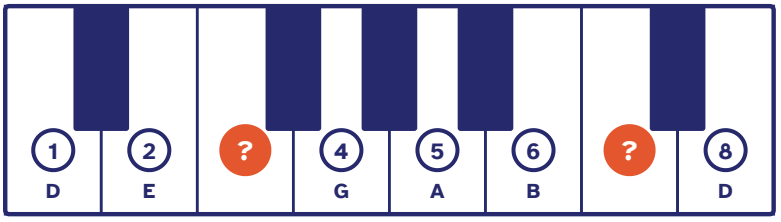



1. **Write the letters** G to G: G A B C D E F G.
2. **Check the top:** 6 to 7 must be T (one key between), 7 to 8 must be S (none between). E to F has nothing between.
3. **Move the ? one key right.** What is it called now?

The **?** becomes **F♯**. G major: G A B C D E F♯ G, one sharp.

EXAMPLE 3

### Build D major

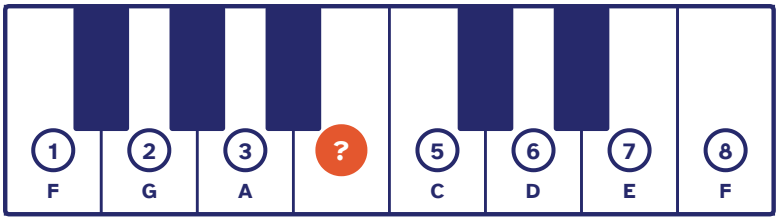



1. **Write the letters** D to D.
2. **Check each step** against T T S T T T S: is there a key between?
3. **Two steps are too short.** Move each ? one key right.

D major: D E **F♯** G A B **C♯** D, two sharps.

EXAMPLE 4

### Build F major



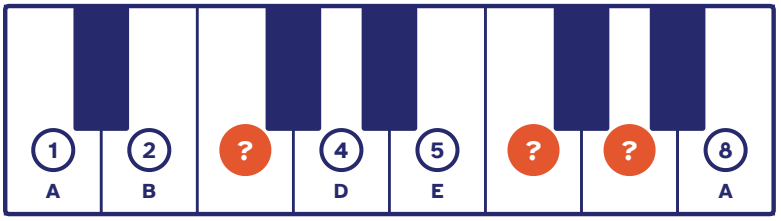

1. **Write the letters** F to F: F G A B C D E F.
2. **Check 3 to 4:** it must be S, nothing between. A to B has a key between: too big.
3. **Move the ? one key left.** What is it called now?

The **?** becomes **B♭**. F major: F G A B♭ C D E F, one flat.

PART IV · SCALES AND KEYS · 1 · CONTINUED

EXAMPLE 5

### Build A major




1 **Write the letters** A to A.

2 **Check each step:** two are too short and the top one is too big.

3 **Move each ? one key right.**

A major: A B **C♯** D E **F♯** **G♯** A, three sharps.

GAME

### Spot the wrong note

**1** G A B C D E F G

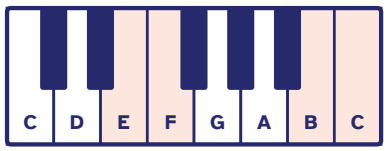
**2** D E F♯ G A B C D

**3** F G A B C D E F

Each of these major scales has one note that breaks T T S T T T S. Which one, and what should it be?

Answer: 1 F should be F♯ · 2 C should be C♯ · 3 B should be B♭.

TRY IT · 10 SECONDS
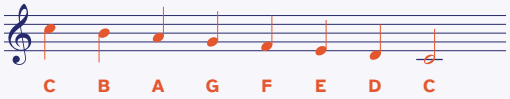



### One finger, C to C.

Play the white keys from C up to C and say “tone, tone, semitone, tone, tone, tone, semitone”. Feel the two short steps.

YOU KNOW THIS

### Joy to the World




The first line is the **C major scale**, top to bottom: C B A G F E D C.

**Try it.** Play C major down from the top C and sing “Joy to the world” over it.

TRAP

- × Using the same letter twice: every letter appears once (F♯ in G major, never G♭).
- × Forgetting the short step at the top: 7 to 8 never has a key between.

SEE ALSO

Minor scales p. 96 · Keys and key signatures p. 99

PRACTISE IT

Workbook p. 1 · Online: Library › Scales

PART IV · SCALES AND KEYS · 2

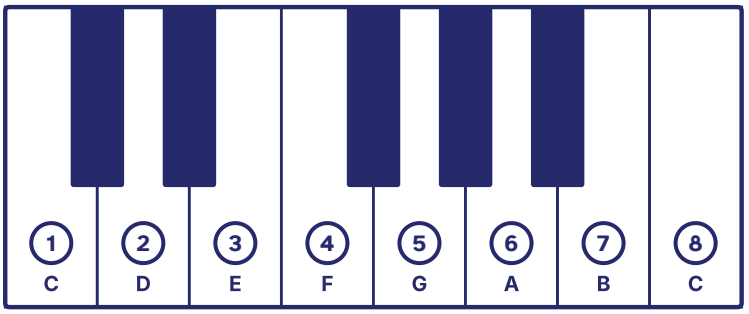
# Minor scales

Start from the major scale on the same key note and move three notes, the **3rd**, **6th** and **7th**, one key down. That is the **natural minor**: same home, darker colour.

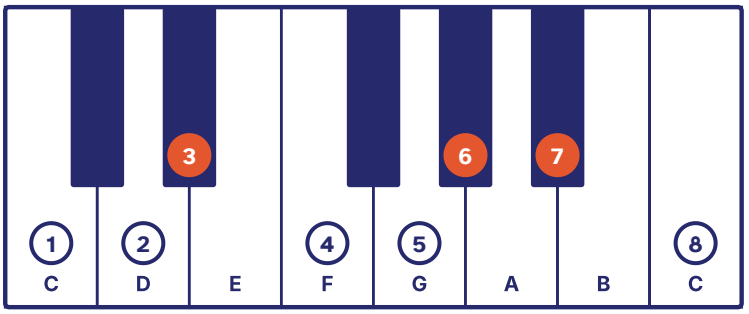
THE SECRET

**Minor = the major on the same note with 3, 6 and 7 one key lower.**

**C major**




**C minor: 3, 6 and 7 move one key left**
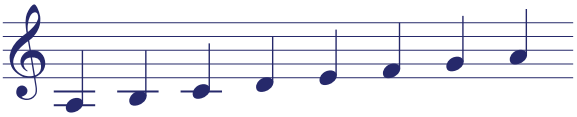



*The same home note · three notes one key lower*

THE RULE
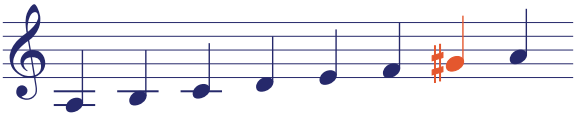
**Natural minor = the major with 3, 6 and 7 lowered one key. Harmonic minor = the natural minor with 7 raised again. Melodic minor = going up, 6 and 7 raised; coming down, the natural minor.**

## A minor in three forms
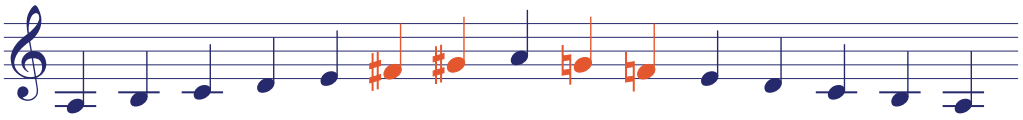


Natural: A major (F♯ C♯ G♯) with 3, 6 and 7 lowered: C, F and G. All white keys.

PART IV · SCALES AND KEYS · 2 · CONTINUED



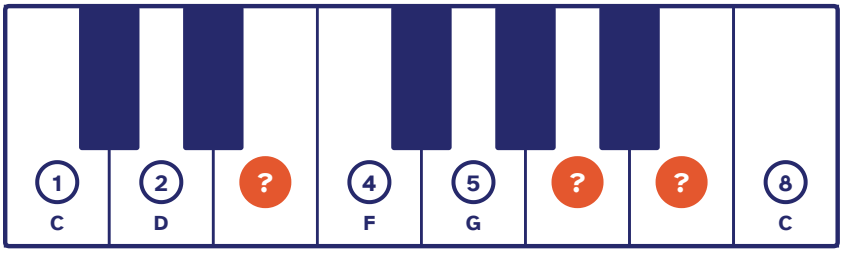
Harmonic: the 7th raised again, one key up: **G♯**.



Melodic: going up, 6 and 7 raised (**F♯ G♯**); coming down, back to **G F**.

EXAMPLE 1

### Build C natural minor
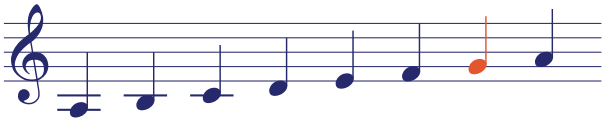



1. **Start from C major:** C D E F G A B C.
2. **The ? keys are 3, 6 and 7.** Move each one key left.
3. **Name the keys you land on.**

C natural minor: C D **E♭** F G **A♭** **B♭** C.

EXAMPLE 2

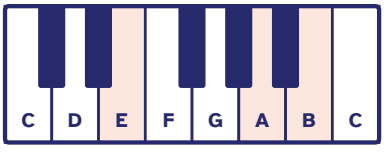
### Build A harmonic minor



1. **Start from A natural minor:** A B C D E F G A, all white.
2. **Take note 7** (the ?) and raise it one key.
3. **Name it.**

Note 7 becomes **G♯**: A harmonic minor is A B C D E F G♯ A.

TRY IT · 10 SECONDS



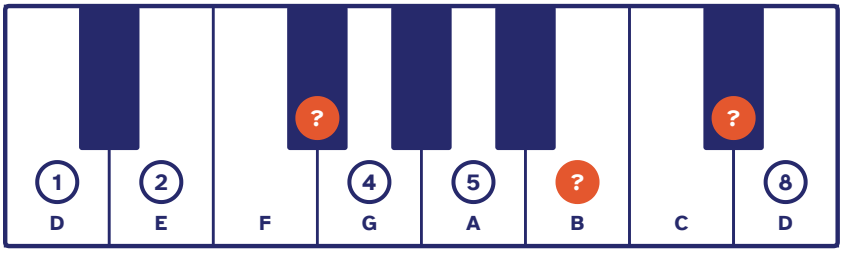

### Major, then minor.

Play C major, then play it again with E, A and B one key lower. Same home, new colour: bright, then dark.

PART IV · SCALES AND KEYS · 2 · CONTINUED

EXAMPLE 3

### Build D natural minor




1 **This is D major.** The ? keys are notes 3, 6 and 7.

2 **Move 3, 6 and 7** one key left each.

3 **Name the new keys.**

D natural minor: D E **F** G A **B♭** **C** D.

GAME

### Which minor?

1

**nothing raised**

2

**the 7th raised**

3
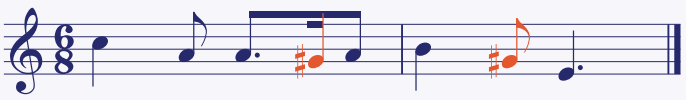
**6th and 7th raised going up, lowered coming down**

Three forms of the minor scale. Name each one.

Answer: 1 natural · 2 harmonic · 3 melodic.

YOU KNOW THIS

### Greensleeves



A minor, with a **G♯**: the raised 7th of the harmonic minor, leaning hard up to A.

**Try it.** Play it once with G♯, once with G: hear the pull disappear.

TRAP

- × Lowering the wrong notes: only 3, 6 and 7 move.
- × Playing the melodic minor's raised notes on the way down.

SEE ALSO
Major scales p. 93 · Keys and key signatures p. 99

PRACTISE IT
Workbook p. 2 · Online: Library › Scales

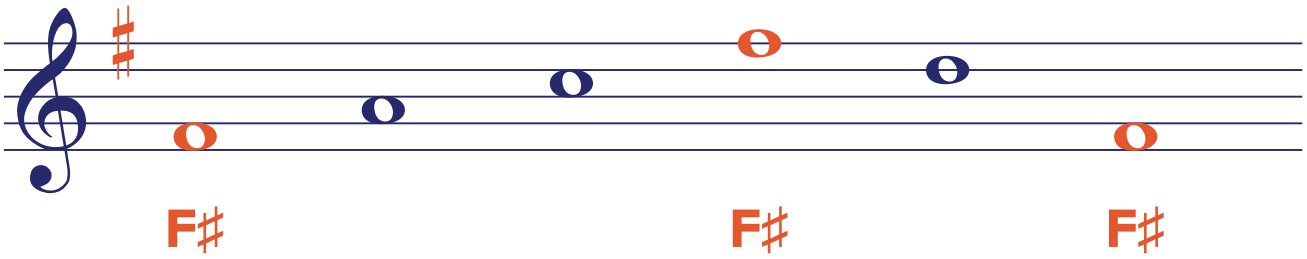
PART IV · SCALES AND KEYS · 3

# Keys and key signatures

Why write F♯ fifty times? The **key signature** at the start of each line says it once, for the whole piece. Its sharps or flats always line up in the same order.

THE SECRET

**The key signature says once, at the start, which notes are always sharp or flat.**




*One sharp at the start: every F plays F♯*



Sharps always come in this order: F C G D A E B.



Flats: B E A D G C F, the sharps backwards.

To remember the order: **F**ather **C**harles **G**oes **D**own **A**nd **E**nds **B**attle; backwards for the flats: **B**attle **E**nds **A**nd **D**own **G**oes **C**harles' **F**ather.

| KEY YOU BUILT | KEY SIGNATURE |
|---|---|
| C major | none |
| G major | 1♯: F♯ |
| D major | 2♯: F♯ C♯ |
| F major | 1♭: B♭ |

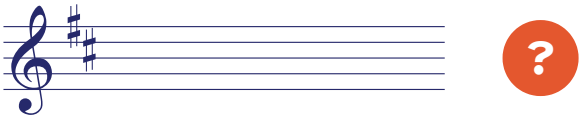
## One signature, two keys

Every key signature also belongs to a **minor** key: the **relative minor**. To find it, go **down a minor 3rd**: 2 keys between, then land. C major and A minor share every note: all white keys.

PART IV · SCALES AND KEYS · 3 · CONTINUED

EXAMPLE 1

### Which notes are sharp?

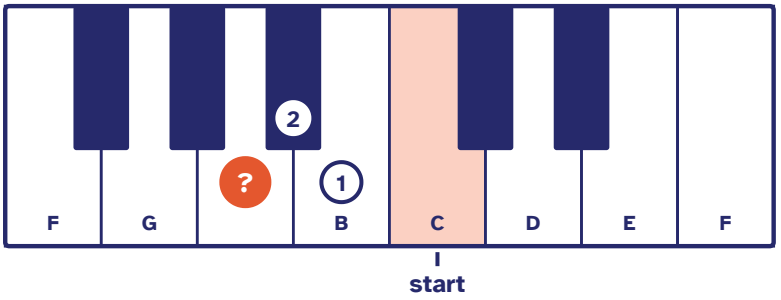



1. **Count the sharps.**
2. **Name them in order:** the first sharp is always F, then ...
3. **Which keys turn black** for the whole piece?

Two sharps, **F♯ and C♯**: every F and every C in the piece, high or low.

EXAMPLE 2

### Find the relative minor of C major

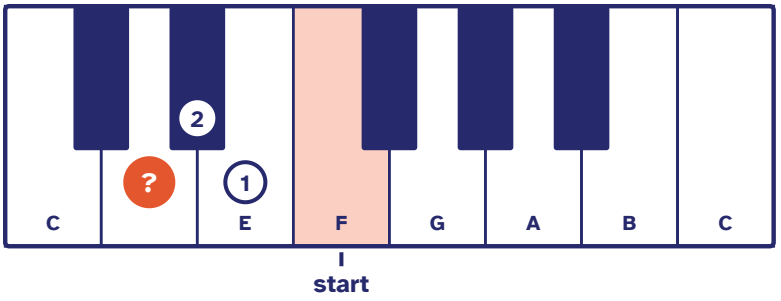



1. **Start on C**, the key note.
2. **Count 2 keys between, going down:** B is 1, B♭ is 2 ...
3. **The next key is the ?.**

You land on **A**: the relative minor of C major is **A minor**.

EXAMPLE 3

### Find the relative minor of F major




1. **Start on F.**
2. **Count 2 keys between, going down:** E is 1, E♭ is 2 ...
3. **The next key is the ?.**

The **?** is **D**: D minor, which shares F major's one flat, B♭.

PART IV · SCALES AND KEYS · 3 · CONTINUED

EXAMPLE 4

## Which notes are flat?



1. **Count the flats.**
2. **Name them in order:** the first flat is always B, then …
3. **Which keys turn black?**

Three flats, **B♭ E♭ A♭**: every B, E and A in the piece.

GAME

## Which comes next?

1 sharps: F C G …

2 flats: B E …

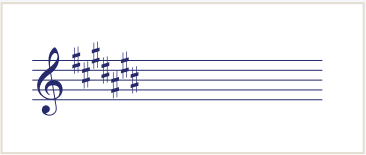
3 the 1st flat

4 the 2nd sharp

Use the orders of sharps and flats.

Answer: 1 D · 2 A · 3 B♭ · 4 C♯.

TRY IT · 10 SECONDS
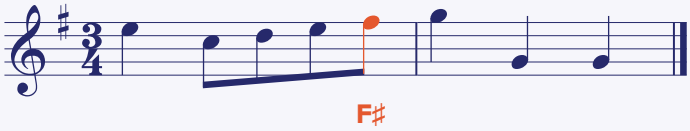


**Say the sharps in order.**

"Father Charles Goes Down And Ends Battle": F C G D A E B. Then backwards for the flats: B E A D G C F.

YOU KNOW THIS

## Minuet in G




One sharp in the key signature: every F is **F♯**, like the one just before the top G. One sharp, and the piece ends on G: **G major**.

**Try it.** Find every F in the piece and play it as F♯.

THINK OF IT A dress code on a party invitation: written once at the top, it applies to every guest all night. Only an accidental, a word to one guest, overrules it, and only until the bar line.

TRAP

- × Writing a key signature in the wrong order: sharps always F C G D A E B.
- × Reading it once and forgetting it: glance at the key signature at the start of every line.

SEE ALSO

Major scales p. 93 · Finding the key p. 102 · The circle of fifths p. 105

PRACTISE IT

Workbook p. 3 · Online: Library › Keys

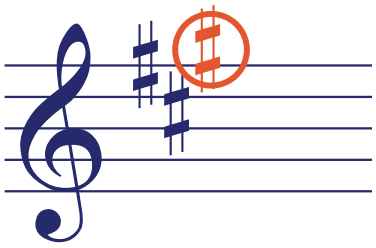
PART IV · SCALES AND KEYS · 4

# Finding the key

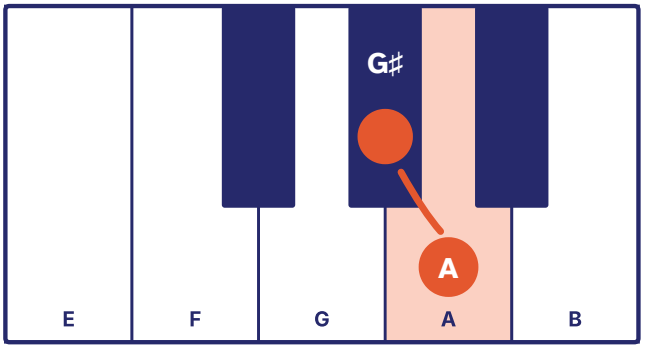
A key signature names two keys, a major and its relative minor. Two quick rules find the major; a minor 3rd down finds the minor; then the music itself tells you which of the two it is.

**THE SECRET**

**Last sharp, then the next key up. Second-last flat: that is the key.**



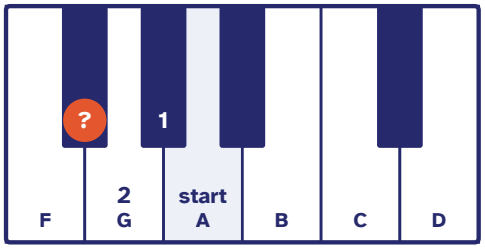
last sharp: G♯




*Sharps: the last sharp, then the very next key up. Three sharps: A major*



**Flats: the second-last flat is the key.**
B♭ E♭ A♭: the second-last is E♭, so **E♭ major**. One flat alone is F major.




**Major to minor: down a minor 3rd.**
From A: 2 keys between, G♯ and G, then land on **F♯**. Three sharps is A major **or** F♯ minor.

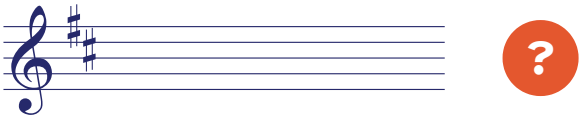
## Minor keys and their clue notes

| MINOR KEY | SIGNATURE | CLUE NOTE (THE RAISED 7TH) |
|---|---|---|
| A minor | none | G♯ |
| E minor | 1♯ | D♯ |
| D minor | 1♭ | C♯ |
| B minor | 2♯ | A♯ |
| G minor | 2♭ | F♯ |

PART IV · SCALES AND KEYS · 4 · CONTINUED

EXAMPLE 1

### Name the major key: sharps



1. **Find the last sharp,** the one furthest right.
2. **Name it** (F first, then C ...).
3. **Go to the very next key up.** That is the key.

The last sharp is C♯; the next key up is D: **D major**.

EXAMPLE 2

### Name the major key: flats
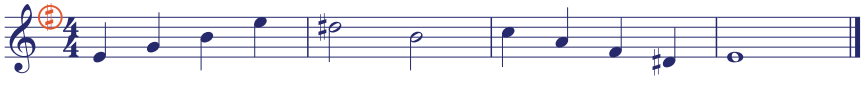


1. **Find the second-last flat.**
2. **Name it** (B first, then E ...).
3. **That flat names the key.**

B♭ E♭: the second-last is B♭: **B♭ major**.

EXAMPLE 3

### Major or minor?



1. **The key signature:** one sharp. Which major, and which minor a minor 3rd below it?
2. **Look for a sharp that is not in the signature.** A minor key raises its 7th: whose 7th is it?
3. **Look at the last note.**

One sharp, D♯ clues, ending on E: **E minor**, not G major.

GAME

### Name the major key

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| 2 sharps | 4 sharps | 1 flat | 3 flats |

Sharps: the last sharp, the very next key up. Flats: the second-last flat (one flat alone is F).

Answer: 1 D · 2 E · 3 F · 4 E♭.

PART IV · SCALES AND KEYS · 4 · CONTINUED

EXAMPLE 4

### Find a minor key from its clue



1. **Two flats:** B♭ major, or the minor a minor 3rd below it.
2. **Count 2 keys between going down** from B♭: A, A♭ … and land. Which minor key is it, and what is its 7th note?
3. **Raise that 7th**: if the piece is full of it, the key is minor.

Two flats with F♯ clues: **G minor** (its 7th, F, raised to F♯).

TRY IT · 10 SECONDS



### Three sharps, stop, step up.

Say “F C G”, stop: the last one is G♯. The next key up is A. Three sharps is A major. Try it with two and four.
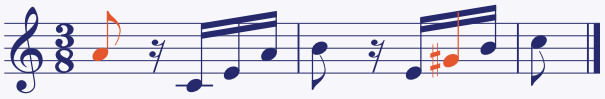
HOW TO

### Find the key of a new piece

1. **The key signature:** name the major key, and its relative minor a minor 3rd below.
2. **The clue notes:** look for the minor key’s raised 7th.
3. **The last bass note:** pieces nearly always end on the key note.

YOU KNOW THIS

### Für Elise (Beethoven)



No key signature, a G♯ as the clue, and the music keeps coming home to A: **A minor**.

**Try it.** Play A, then G♯ A: hear the G♯ lean into home.

THINK OF IT Detective work: the key signature narrows it to two suspects; the clue note and the last note name the culprit.

SEE ALSO
Keys and key signatures p. 99 · The circle of fifths p. 105

PRACTISE IT
Workbook p. 4 · Online: Library › Keys

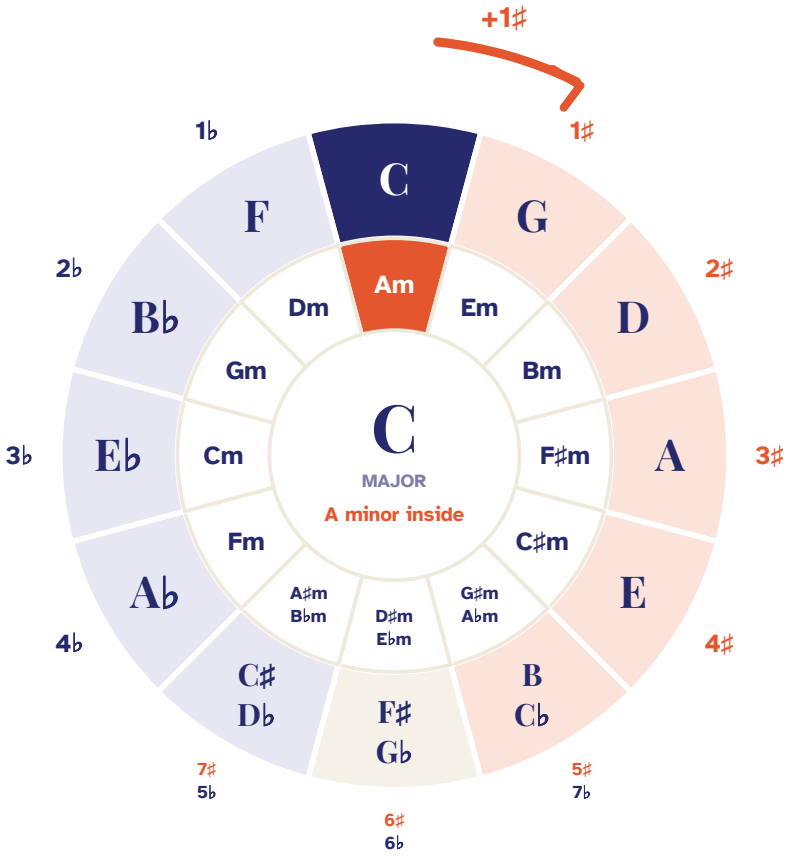
PART IV · SCALES AND KEYS · 5

# The circle of fifths

All the keys on one clock face. Go **right** round the circle and each key has **one more sharp**. Go **left** and each key has **one more flat**.

THE SECRET

**Go right: one more sharp. Go left: one more flat.**




*Right: one more sharp · left: one more flat*

## How to read the clock

**Outer ring:** the major keys. **Inner ring:** their relative minors. **Outside the circle:** how many sharps or flats.

## Neighbours are close

Keys side by side share all but one note, so music moves between them easily: from C major to G major or F major, or to the relative minor.

THE RULE

**Each step right is a 5th up: count the notes, C D E F G, and G is the next key round. That is why it is called the circle of fifths.**

PART IV · SCALES AND KEYS · 5 · CONTINUED

## The first steps round the circle

| STEPS | RIGHT (SHARPS) | LEFT (FLATS) |
|---|---|---|
| 1 | G major: F♯ | F major: B♭ |
| 2 | D major: F♯ C♯ | B♭ major: B♭ E♭ |
| 3 | A major: F♯ C♯ G♯ | E♭ major: B♭ E♭ A♭ |

EXAMPLE 1

### How many sharps in A major?




1. **Start at C**, at the top: no sharps.
2. **Go right** one step at a time: G, D …
3. **Count the steps** to A.

Three steps right: A major has **three sharps**, F♯ C♯ G♯.

EXAMPLE 2

### How many flats in E♭ major?

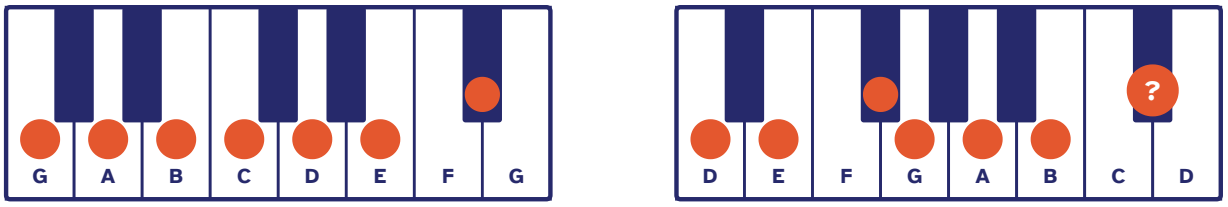



1. **Start at C** again.
2. **Go left:** F, B♭ …
3. **Count the steps** to E♭.

Three steps left: E♭ major has **three flats**, B♭ E♭ A♭.

EXAMPLE 3

### Why do G and D sound so close?



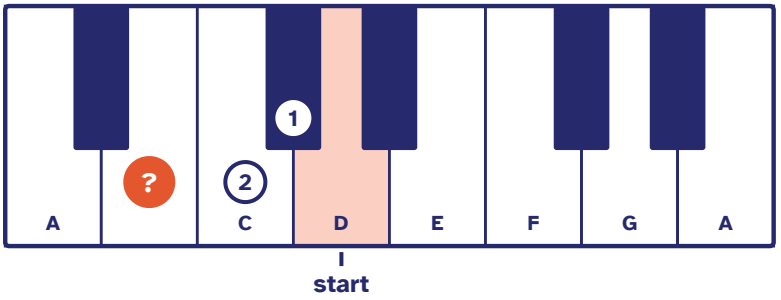

1. **G major** (left) and **D major** (right) are neighbours on the clock.
2. **Compare their notes:** which letters are the same?
3. **Which one letter changes?**

Six notes are the same; only C becomes **C♯**. One step round the circle changes just one note.

PART IV · SCALES AND KEYS · 5 · CONTINUED

EXAMPLE 4

## Find the minor inside D




1. **Start on D**, on the outer ring.
2. **Count 2 keys between, going down:** C♯ is 1, C is 2 …
3. **The next key is the ?.** It sits on the inner ring, just inside D.

The **?** is **B**: B minor, D major's relative minor. Both have two sharps.

GAME

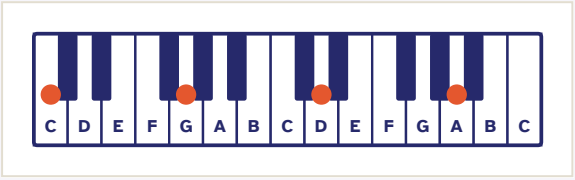
## Neighbours on the clock

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| right of D | left of C | opposite C | inside G |

Right adds a sharp. Inside each major sits its relative minor.

Answer: 1 A · 2 F · 3 F♯ / G♭ · 4 E minor.

TRY IT · 10 SECONDS




### Walk the circle: C G D A.

Play each key note with one finger, a 5th up each time. You just walked three steps right round the circle.

HOW TO

## Learn the circle by heart

1. **Right, out loud:** C G D A E B F♯.
2. **Left:** C F B♭ E♭ A♭ D♭ G♭.
3. **Check yourself:** the number of steps from C is the number of sharps or flats.

TRAP

- × Going the wrong way round: right adds sharps, left adds flats.
- × Forgetting the relative minors on the inner ring.

SEE ALSO

Keys and key signatures p. 99 · Finding the key p. 102 · Chords in a key p. 116

PRACTISE IT

Workbook p. 5 · Online: Library › Related keys

PART IV · SCALES AND KEYS · TIPS

# When a note sounds wrong

You read the note right, yet it sounds sour. The slip is usually in the key, not the note. Name the slip first, then fix only that note.
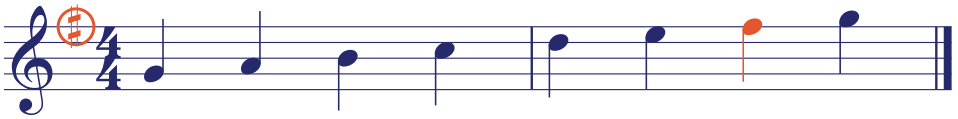
TIP

### What you notice, what to check

| WHAT YOU NOTICE | CHECK | TURN TO |
|---|---|---|
| A note sounds sour, though you read it right | **the key signature**: is this note sharp or flat all the way through? | p. 99 |
| A key with many sharps or flats feels lost | **the circle**: go right for a sharp, left for a flat | p. 105 |
| A minor tune sounds wrong near the end | **the raised 7th**: harmonic and melodic minor raise it | p. 96 |
| A scale sounds wrong in the middle | **the short steps** (nothing between): 3–4 and 7–8 in every major scale | p. 93 |
| A sharp is missing from the key signature | **the order of sharps**: F C G D A E B | p. 99 |
| You are not sure which major key it is | **the last sharp**: the key is the very next key up | p. 102 |

EXAMPLE

### Find the slip



1. **Read the key signature:** one sharp, on the top line, F (ringed).
2. **The red note is an F**, so the sharp belongs to it too: every F, high or low.
3. **No natural cancels it**, so this F is F♯, here and all through the piece.

Play **F♯**, the black key. Plain F is the most common slip in G major.

HOW TO

### Three checks before you play

1. **The key signature:** say its sharps or flats aloud.
2. **The key:** name it from the last sharp or the second-last flat, then check the last note.
3. **The scale:** play it once, so your fingers know which black keys belong.

THINK OF IT Like proofreading: a spelling mistake jumps out once you know which words you usually get wrong. Keep a short list of your own slips, and check those first.